Project Proposal: IoT-Based Temperature Monitoring System for Greenhouses Using Artix-7 FPGA and ESP32

Project Title: IoT-Based Temperature Monitoring System for Greenhouses Using Artix-7 FPGA and ESP32 with Adafruit IO Integration

Project Overview:

The aim of this project is to develop an Internet of Things (IoT)-based greenhouse monitoring system that tracks ambient temperature and provides visual feedback through an RGB OLED screen. The system utilizes the Artix-7 FPGA for high-speed sensor data processing and an ESP32 for wireless cloud integration. Instead of a mechanical alarm, the system employs a dynamic visual alert system: the OLED display background or text color changes to Red if temperatures exceed a safety threshold and remains Green during optimal conditions.

Objectives:

- **Precision Measurement:** Measure greenhouse temperature using the PMOD TMP3 sensor via I2C protocol.
- **Visual Alert System:** Implement a color-coded feedback loop on the PMOD OLED RGB where display colors represent climate status (Green = Safe, Red = Alert).
- **Remote Telemetry:** Transmit real-time data via the ESP32 to the Adafruit IO platform for web-based monitoring.
- **Hardware Acceleration:** Utilize the Artix-7 FPGA to manage timing-critical display drivers and sensor polling.

Components Used:

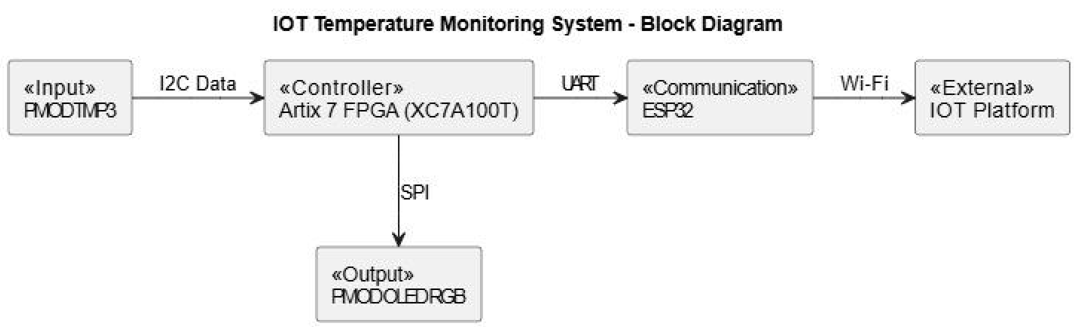
- **Artix-7 100T FPGA:** The primary controller responsible for sensor interfacing and display logic.
- **PMOD TMP3:** Digital temperature sensor for accurate climate monitoring.
- **PMOD OLED RGB:** A 96x64 RGB display used for real-time data visualization and status signaling.
- **ESP32:** Wi-Fi-enabled microcontroller used as a communication bridge between the FPGA and the Cloud.
- **Adafruit IO:** The IoT cloud service for data logging and remote mobile dashboarding.

Methodology:

1. **Sensor Integration:** The Artix-7 FPGA interfaces with the PMOD TMP3 to fetch temperature data.

2. **Logic-Based Alert System:** The FPGA compares the incoming temperature value against a predefined setpoint.
    - If Temp < Threshold, the OLED driver is commanded to render text/background in **Green**.
    - If Temp > Threshold, the OLED driver switches the UI to **Red** to signal an overheat condition.
3. **Wireless Transmission:** The FPGA sends the temperature string to the ESP32 via UART communication. The ESP32 then connects to the local network to push data to the Adafruit IO dashboard.
4. **Remote Monitoring:** Users can view the live temperature and historical logs on via the Adafruit IO interface.

System Block Diagram:



Expected Outcome:

The final system will provide a seamless monitoring experience where greenhouse conditions are visible at a glance. By replacing a buzzer with an RGB OLED alert, the system provides a non-intrusive yet highly effective visual warning system, backed by robust IoT connectivity for remote management.